Status	Finished
Started	Thursday, 29 October 2026, 4:42 PM
Completed	Thursday, 29 October 2026, 4:48 PM
Duration	6 mins 33 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

a. Being the most efficient sorting algorithm
b. Using the divide-and-conquer approach
c. Sorting data in a non-sequential manner
d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **2**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 80,150,200,470,40,90,300,400,70,120
c. 200,470,80,150,40,90,300,400,70,120
d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Extraction
c. Insertion
d. Selection

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Merge Sort
b. Linear Search
c. Bubble Sort
d. Quick Sort

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It compares adjacent elements and swaps them if necessary
b. It divides the list into two halves, sorts each half, and then merges them
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Binary Search
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is less efficient than bubble sort
b. It does not require recursion
c. It is based on the divide-and-conquer approach
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Quick sort
c. Merge sort
d. Heap sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Bubble Sort
- b. Merge Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **10**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It makes the analysis slower
- c. It complicates the analysis process
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **11**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It slows down the process
b. It makes the data unsorted
c. It complicates the selection of items
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Provides a basis for other algorithms to work efficiently
c. Makes data harder to manage
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Linear Search
c. Bubble Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Merge Sort